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Multidrug-resistant *Escherichia coli* in selected sachet water and associated public health risk assessment in Ago-Iwoye and Oru, Ogun State, Nigeria

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Abstract

Safe drinking water remains essential for public health, particularly in developing regions where sachet water is widely consumed. This study assessed the physicochemical properties, bacteriological quality, antibiotic resistance, heavy metal contamination, and potential health risks of 8) selected sachet water samples. Physicochemical parameters were measured using APHA standard methods. Bacterial counts were determined by serial dilution and plate count techniques. *Escherichia coli* isolates were identified through colonial morphology, Gram staining, and biochemical tests. Antibiotic susceptibility was assessed by Kirby–Bauer disc diffusion, while MAR index, heavy metals, correlations, CF, HPI, and HRI were calculated. Physicochemical parameters showed moderate variation, with pH ranging from 5.80 to 7.37, EC (0.20–1.20 mS/cm), TDS (100.00–600.00 ppm), turbidity (0.90–1.90 NTU), and DO (6.10–7.20 mg/L). Organic pollution indicators were highest in W3 (BOD: 1.80 mg/L, COD: 8.50 mg/L, TOD: 11.50 mg/L). Total heterotrophic counts ranged from 0.00 to 110.00 CFU/ml, exceeding permissible limits, while coliform counts peaked at 6.00 CFU/ml. Isolates identified as *Escherichia coli* exhibited high susceptibility to ofloxacin (26.00–34.00 mm) but complete resistance (0.00 mm) to several antibiotics. MAR indices ranged from 0.63 to 0.75, indicating high resistance. Heavy metals were within permissible limits, with Pb up to 0.006 ppm and Fe up to 0.160 ppm. Risk indices confirmed low contamination (CF < 1; HPI: 34.65–52.15; HRI: 0.159–0.220). Despite acceptable chemical quality, bacteria contamination and multi-drug resistance pose significant public health concerns.

Keywords Bacteria, *Escherichia coli*, Health risks, Multi-drug resistance, Sachet water



1 Introduction

The consumption of safe drinking water is essential for maintaining public health and preventing the spread of waterborne diseases. In many developing countries, including Nigeria, packaged drinking water such as sachet water has become a widely consumed alternative to municipal water supplies due to its affordability and accessibility [1, 2]. Sachet water, often referred to as “pure water,” is commonly sold by street vendors and small-scale producers across urban and rural communities [3]. However, concerns have been raised regarding the microbiological and physicochemical quality of these products, particularly where production and distribution are not adequately monitored by regulatory authorities such as the National Agency for Food and Drug Administration and Control [4].

One of the major bacterial contaminants associated with drinking water is *Escherichia coli*, which belongs to the family *Enterobacteriaceae*, a Gram-negative bacterium commonly found in the intestinal tract of humans and warm-blooded animals [5, 6]. The presence of this organism in drinking water is widely used as an indicator of faecal contamination and poor sanitary conditions during water processing, storage, or handling [7]. Contamination of sachet water with *Escherichia coli* may occur through untreated water sources, unhygienic production practices, or improper storage conditions [8, 9]. Consumption of water contaminated with pathogenic strains of *Escherichia coli* can lead to gastrointestinal infections, diarrhea, urinary tract infections, and other serious health complications [10].

In recent years, the emergence of multidrug-resistant (MDR) bacteria in environmental and drinking water sources has become a major global public health concern [11–13]. Multidrug resistance occurs when bacterial strains develop resistance to multiple classes of antibiotics, thereby reducing the effectiveness of commonly used antimicrobial therapies [14]. Among waterborne bacteria, antibiotic-resistant *Escherichia coli* is particularly significant because it serves as both an indicator of faecal contamination and a reservoir of antimicrobial resistance genes that can be transferred to other microorganisms through horizontal gene transfer [15]. Environmental water systems have increasingly been recognized as important reservoirs of the environmental resistome, facilitating the persistence, evolution, and dissemination of antimicrobial resistance genes among bacterial populations. The continuous exposure of environmental bacteria to antibiotic residues, heavy metals, and other pollutants may promote co-selection mechanisms that enhance the spread of resistance determinants, thereby contributing significantly to the global antimicrobial resistance crisis recognized by the World Health Organization [16].

The contamination of drinking water with MDR *Escherichia coli* poses substantial health risks, particularly in developing countries where sanitation infrastructure, water treatment systems, and regulatory oversight may be inadequate [17]. In Nigeria, sachet water has become one of the most accessible and affordable sources of potable water due to deficiencies in public water supply systems [18]. Although sachet water consumption has improved access to drinking water, concerns remain regarding bacterial contamination, physicochemical quality, and the presence of antibiotic-resistant bacteria in commercially available products [19]. Previous studies in Nigeria have documented variations in physicochemical and bacteriological water quality across different locations and seasons. Studies conducted in Awka, southeastern Nigeria, reported significant fluctuations in bacterial loads and physicochemical parameters in drinking water

sources, emphasizing the need for continuous surveillance and quality assessment [20, 21]. Despite these findings, there remains a paucity of data regarding the occurrence and antibiotic resistance profiles of *Escherichia coli* isolated from sachet water sold in Ago-Iwoye and Oru communities of Ogun State, southwestern Nigeria. This knowledge gap limits effective risk assessment and evidence-based interventions aimed at protecting public health in these rapidly growing communities.

Beyond bacterial contamination, water quality assessments are increasingly being framed within the broader context of both legacy pollutants and emerging contaminants. While traditional concerns such as heavy metals remain important, recent studies have highlighted the growing threats posed by contaminant mixtures including microplastics, pharmaceutical residues, per- and polyfluoroalkyl substances (PFAS), and other emerging pollutants in groundwater and drinking water systems worldwide [22, 23]. Recent reviews have emphasized that heavy metal contamination remains a sidelined but persistent environmental challenge despite the increasing attention given to emerging contaminants [24, 25]. Furthermore, groundwater systems are increasingly threatened by complex contaminant cocktails involving microplastics, chemical pollutants, and resistant microorganisms that may interact synergistically to affect ecosystem and human health [24]. Emerging contaminants such as PFAS have also been identified as persistent pollutants capable of compromising groundwater quality and posing long-term health risks even at low concentrations [26].

The presence of heavy metals and other physicochemical contaminants in drinking water may further exacerbate environmental and human health risks. Heavy metals such as lead, cadmium, nickel, and iron can enter water systems through natural geological weathering processes and anthropogenic activities including agricultural runoff, industrial discharge, and improper waste disposal [27]. Chronic exposure to elevated concentrations of these metals has been associated with neurological disorders, kidney dysfunction, developmental abnormalities, carcinogenic effects, and other adverse health outcomes [28]. Moreover, increasing evidence suggests that heavy metals may contribute to the selection and maintenance of antibiotic resistance genes in environmental microbial communities through co-selection mechanisms, thereby linking chemical pollution with antimicrobial resistance development [29].

Given the widespread consumption of sachet water in Ago-Iwoye and Oru and the limited information available on its bacteriological safety, physicochemical quality, heavy metal burden, and antibiotic resistance characteristics, this study was designed to evaluate the occurrence and multidrug resistance patterns of *Escherichia coli* isolates recovered from selected sachet water brands. The findings will contribute to filling existing knowledge gaps regarding drinking water quality in southwestern Nigeria and provide baseline data for public health monitoring, regulatory enforcement, and sustainable water resource management.

Risk assessment approaches are widely used to evaluate the potential health implications of contaminated water [30]. These approaches include the determination of contamination indices, heavy metal pollution indices, and health risk indices, which help quantify the level of exposure and potential toxicity associated with environmental contaminants [31]. When combined with antibiotic susceptibility testing, these assessments provide valuable insight into the potential public health threats associated with contaminated drinking water, particularly in communities that rely heavily on packaged water

for daily consumption [32]. Therefore, this study aimed to investigate the multidrug resistance profile of *Escherichia coli* isolated from selected sachet water samples sold in Ago-Iwoye and Oru environments in Ogun State, Nigeria. The study also evaluated the potential health risks associated with heavy metal contamination in these water samples.

2 Materials and methods

2.1 Study area

This study was conducted in Ago-Iwoye, Oru, and Awa communities located within Ijebu North Local Government Area of Ogun State, southwestern Nigeria. Ogun State lies within the tropical rainforest zone and experiences distinct wet and dry seasons, which may influence water quality characteristics. Ago-Iwoye is a rapidly urbanizing university town that hosts a large transient and resident population due to the presence of Olabisi Onabanjo University, commercial establishments, and growing residential developments [33]. Oru is a neighbouring semi-urban community characterized by residential settlements, markets, educational institutions, and commercial activities [33, 34]. Both communities depend heavily on commercially packaged sachet water as a major source of drinking water due to increasing population growth and variable access to public water supplies. The widespread consumption of sachet water in these communities makes them suitable locations for assessing physicochemical quality, heavy metal contamination, bacteriological status, and antimicrobial resistance patterns associated with packaged drinking water.

2.2 Sample collection

A preliminary market survey was conducted to identify the sachet water brands most commonly available and consumed within Ago-Iwoye and Oru communities. Based on the frequency of availability across major distribution points, 8 sachet water brands were selected for analysis (Fig. 1). The selected brands represented the most frequently encountered products among vendors, retail shops, and street hawkers at the time of sampling and were considered representative of the dominant brands consumed by residents in the study area. Sachet water samples were purchased in their original sealed condition to avoid external contamination in February, 2026. Each sample was carefully labelled according to location and brand before transportation to the laboratory. Samples were transported in sterile containers using ice-packed coolers and analysed within 24 h of collection to preserve sample integrity and ensure reliable bacteriological results. Standard aseptic procedures were maintained throughout sample handling and processing.

2.3 Study limitation

Although several sachet water brands are distributed within Ago-Iwoye and Oru, only eight widely available brands were included in this study due to logistical and resource constraints. Consequently, the findings may not fully represent the quality characteristics of all sachet water brands marketed within the study area.

2.4 Physicochemical analysis of sachet water samples

The physicochemical parameters of the water samples were determined using standard procedures recommended by [35, 36]. Colour, odour, and taste were assessed

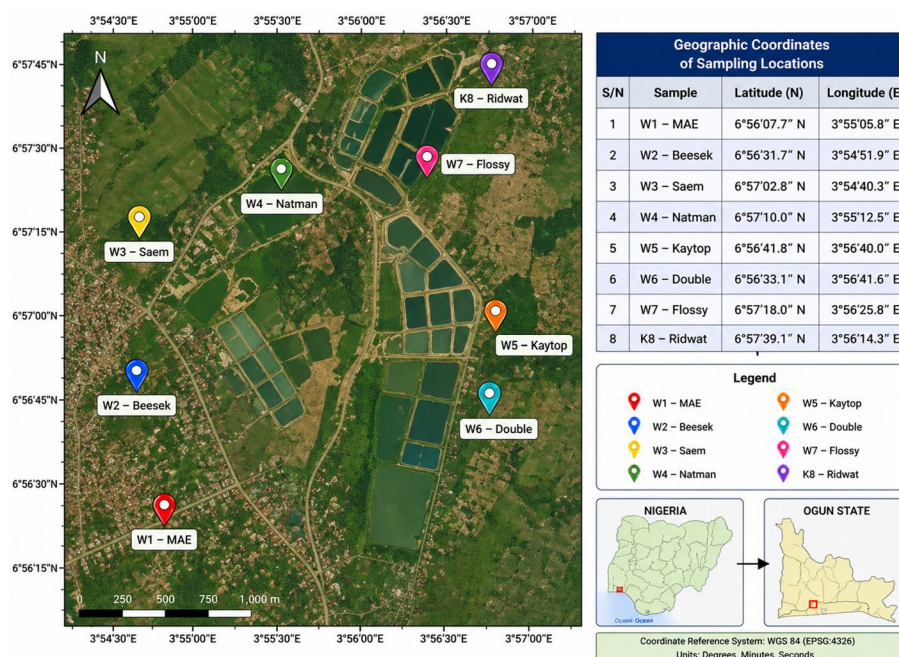


Fig. 1 Geographic coordinates of sampling locations

organoleptically by visual observation and sensory evaluation immediately after sample collection. The pH and temperature were measured in situ using a digital pH meter (Hanna Instruments HI98107 pH Meter, Hanna Instruments, Romania). Electrical conductivity (EC) and total dissolved solids (TDS) were determined using a conductivity/TDS meter (Hanna Instruments HI98303 DiST 3 Conductivity/TDS Meter, Hanna Instruments, Romania). Turbidity was measured using a turbidity meter (Hach 2100Q Portable Turbidimeter, Hach Company, USA). Dissolved oxygen (DO) was determined using a portable DO meter (Hanna Instruments HI9146 Dissolved Oxygen Meter, Hanna Instruments, Romania). Hydrogen sulfide (H_2S) concentration was determined using a spectrophotometric method with a UV-Visible spectrophotometer (Shimadzu UV-1800 Spectrophotometer, Shimadzu Corporation, Japan). All measurements were conducted in triplicate to ensure accuracy and reliability of the results.

2.4.1 Determination of biochemical oxygen demand (BOD)

Biochemical Oxygen Demand (BOD_5) was determined using the standard 5-day incubation method according to the procedures described by the American Public Health Association [37, 38]. Water samples were collected in sterile BOD bottles and analyzed immediately to prevent changes in oxygen concentration. The initial dissolved oxygen (DO_i) concentration of each sample was measured using a calibrated dissolved oxygen meter (Hanna Instruments HI9146 Dissolved Oxygen Meter, Hanna Instruments, Romania). Where necessary, samples were diluted with oxygen-saturated dilution water containing phosphate buffer, magnesium sulfate, calcium chloride, and ferric chloride solutions to provide suitable conditions for microbial activity. The bottles were then incubated in the dark at $20 \pm 1^\circ\text{C}$ for five days using a BOD incubator (Memmert BE500 BOD Incubator, Memmert GmbH + Co. KG, Germany) to prevent photosynthetic oxygen production. After incubation, the final dissolved oxygen concentration (DO_f) was

measured. All analyses were performed in triplicate to ensure accuracy and reliability [39].

2.4.2 Determination of chemical oxygen demand (COD)

Chemical Oxygen Demand (COD) was determined using the closed reflux colorimetric method following Standard Methods for the Examination of Water and Wastewater [38]. Approximately 2 mL of homogenized water sample was transferred into a COD digestion vial containing a standardized acidic potassium dichromate digestion solution, silver sulfate catalyst, and mercuric sulfate for chloride interference suppression. The vials were sealed and digested at 150 °C for two hours in a COD reactor (Hach DRB200 COD Reactor, Hach Company, USA). During digestion, oxidizable organic compounds present in the sample were chemically oxidized by potassium dichromate under strongly acidic conditions. Following digestion, the vials were allowed to cool to room temperature and analyzed calorimetrically using a spectrophotometer according to the manufacturer's instructions. COD concentrations were determined by comparing absorbance values against calibration standards prepared from potassium hydrogen phthalate solutions. Results were expressed as mg O₂/L. COD represents the total oxygen equivalent required for the chemical oxidation of both biodegradable and non-biodegradable organic matter present in water. All analyses were performed in triplicate to ensure accuracy and reliability [37, 40].

2.4.3 Determination of total oxygen demand (TOD)

Total Oxygen Demand (TOD) was determined using a Hach BioTector B3500 Analyzer (Hach Company, USA) based on the catalytic combustion oxidation principle. Prior to analysis, water samples were homogenized and filtered where necessary to remove large suspended particles that could interfere with the analytical system. A measured volume of sample was automatically introduced into the analyzer, where all oxidizable substances, including organic and inorganic compounds, were subjected to high-temperature catalytic oxidation in the presence of oxygen. During oxidation, the analyzer continuously measured the amount of oxygen consumed by the sample. The difference between the oxygen concentration entering and leaving the reaction chamber was used to calculate the total oxygen demand. Results were automatically generated by the instrument and expressed as mg O₂/L. Unlike BOD and COD, which estimate biodegradable and chemically oxidizable fractions respectively, TOD measures the total oxygen requirement for complete oxidation of all oxidizable constituents in a sample [41].

2.5 Bacteriological analysis of water samples using plate count agar (PCA) and eosin methylene blue agar (EMB)

The bacteriological quality of the water samples was determined using the standard plate count method [42]. Prior to analysis, all culture media, glassware, and equipment were sterilised using an autoclave at 121 °C for 15 min. For the total heterotrophic bacterial count, Plate Count Agar (PCA) and Eosin Methylene Blue (EMB) agar were used as the culture medium. Approximately 1 mL of the direct water sample was aseptically transferred into sterile Petri dishes, after which molten PCA and EMB agar were poured into the plates using the pour plate technique. The plates were gently swirled to ensure even distribution of the inoculum and allowed to solidify. The inoculated plates were then

incubated at 37 °C for 24–48 h. After incubation, visible bacterial colonies were counted and recorded. The number of colonies observed was expressed as colony-forming units per milliliter (CFU/ml) of the water sample. On EMB agar, *Escherichia coli* typically produces colonies with a metallic green sheen, while other coliform bacteria may produce pink or purple colonies. These colonial characteristics helped in the preliminary identification of potential coliform organisms present in the water samples [43].

2.6 Gram staining and biochemical identification of bacteria

Bacterial isolates obtained from the culture plates were further identified using Gram staining and biochemical tests. Gram staining was performed to determine the Gram reaction and morphological characteristics of the bacterial cells. A smear of each bacterial isolate was prepared on a clean glass slide and heat-fixed. The smear was then stained using crystal violet, followed by the addition of Gram's iodine, which acted as a mordant. The slide was then decolorized using alcohol and counterstained with safranin. The stained slides were examined under a light microscope using oil immersion (100× objective lens) to observe the cellular morphology and Gram reaction. Gram-positive bacteria appeared purple, while Gram-negative bacteria appeared pink or red [44]. Following Gram staining, further identification of the bacterial isolates was carried out using standard biochemical tests. These included the catalase test, oxidase test, indole test, citrate utilization test, methyl red test, Voges–Proskauer test, and sugar fermentation tests. Each test was performed according to standard bacteriological procedures [45].

2.7 Antibiotic susceptibility test of *Escherichia coli* isolates

The antibiotic susceptibility profiles of the *Escherichia coli* isolates recovered from sachet water samples were determined using the Kirby–Bauer disk diffusion method on Mueller–Hinton agar following the guidelines of the Clinical and Laboratory Standards Institute [46, 47]. Pure bacterial isolates were subcultured on nutrient agar and incubated at 37 °C for 18–24 h to obtain fresh cultures. A bacterial suspension was prepared in sterile normal saline and adjusted to the turbidity of a 0.5 McFarland standard (approximately 1.5×10^8 CFU/mL). Sterile cotton swabs were used to evenly inoculate the surface of Mueller–Hinton agar plates. Commercial antibiotic discs (Oxoid, UK) comprising ciprofloxacin (5 µg), gentamicin (10 µg), ceftriaxone (30 µg), amoxicillin–clavulanic acid (20/10 µg), azithromycin (15 µg), cefuroxime (30 µg), erythromycin (15 µg), and vancomycin (30 µg) were aseptically placed on the inoculated agar surface. The plates were incubated aerobically at 37 °C for 18–24 h, after which the diameters of inhibition zones were measured in millimetres using a transparent ruler. The inhibition zone diameters were interpreted according to the breakpoint criteria of Clinical and Laboratory Standards Institute [46]. Isolates tested against ciprofloxacin (5 µg) were classified as susceptible at inhibition zones ≥ 26.00 mm, intermediate at 22.00–25.00 mm, and resistant at ≤ 21 mm. For gentamicin (10 µg), susceptibility, intermediate susceptibility, and resistance were defined as ≥ 15.00 mm, 13.00–14.00 mm, and ≤ 12.00 mm, respectively. Ceftriaxone (30 µg) results were interpreted as susceptible at ≥ 26.00 mm, intermediate at 23.00–25.00 mm, and resistant at ≤ 22.00 mm. For amoxicillin–clavulanic acid (20/10 µg), inhibition zones of ≥ 18 mm were considered susceptible, 14.00–17.00 mm intermediate, and ≤ 13.00 mm resistant. Cefuroxime (30 µg) susceptibility was defined as ≥ 23.00 mm,

intermediate susceptibility as 15.00–22.00 mm, and resistance as ≤ 14.00 mm. Although azithromycin (15 μg), erythromycin (15 μg), and vancomycin (30 μg) were included in the antibiotic panel [46], does not provide routine interpretive criteria for erythromycin and vancomycin against *Escherichia coli* because members of the *Enterobacterales* exhibit intrinsic resistance to these agents. Similarly, azithromycin breakpoints are limited to specific clinical applications and selected *Enterobacterales* species. Therefore, the zone diameters obtained for these antibiotics were recorded and reported for comparative purposes only.

2.8 Heavy metal analysis using atomic absorption spectrophotometry (AAS)

Heavy metal concentrations in the water samples are Cu, Zn, Pb, Ni, Co, Fe, and Cd were determined using Atomic Absorption Spectrophotometry following standard methods [48]. Water samples were first acid-digested with concentrated nitric acid and filtered prior to analysis. Quantification of the metals was carried out using a Buck Scientific Model 210 VGP Atomic Absorption Spectrophotometer manufactured by Buck Scientific, USA. The instrument was calibrated with appropriate standard solutions prepared for each metal, and absorbance readings were taken at their respective wavelengths. Metal concentrations were expressed in parts per million (ppm), and all analyses were performed in triplicate to ensure accuracy and reproducibility of the results.

2.9 Potential risk assessment analysis (PRA)

Potential Risk Assessment (PRA) was determined using Contamination Factor (CF), Heavy Metal Pollution Index (HPI), and Health Risk Index (HRI) [49]. After heavy metal concentrations in water samples were determined and evaluated against World Health Organization (WHO) permissible standards, the Contamination Factor (CF) was calculated for each metal at every sampling site using the formula $CF = C_m/C_s$, where C_m represents the measured metal concentration and C_s denotes the WHO standard value. This index quantifies the degree to which individual metal concentrations deviate from acceptable limits, with values classified as low ($CF < 1$), moderate (1–3), considerable (3–6), or very high (≥ 6).

The Heavy Metal Pollution Index (HPI) was computed to provide an aggregate measure of overall water quality with respect to multiple metals simultaneously. Unit weightages (W_i) were assigned to each metal as the inverse of its WHO standard, normalised so that the sum of all weightages equalled unity. The HPI was then derived as $HPI = \sum(W_i \times C_i/S_i) \times 100$, with values below 100 indicating low pollution and values at or above 100 signifying high pollution.

The Health Risk Index (HRI) was employed to assess potential non-carcinogenic risk to adult consumers (Table 1). HRI was calculated as $HRI = (C \times IR \times EF \times ED) / (BW \times AT \times RfD)$, assuming a daily water intake of 2 L, body weight of 70 kg, exposure frequency of 365 days per year, exposure duration of 70 years, and an averaging time of 25,550 days. Reference dose (RfD) values were sourced from established toxicological databases. An HRI below 1.0 indicates acceptable risk, while values at or exceeding 1.0 denote potential health concern.

Table 1 Input assumption parameters for intake value and reference dose (RfD) in health risk assessment of heavy metals from sachet water [50]

Parameter	Symbol	Assumed value	Unit	Description/basis
Metal concentration	C	Measured value	mg/L	Concentration of heavy metals (Cu, Zn, Pb, Ni, Co, Fe, Cd) detected in sachet water samples
Ingestion rate	IR	2.0	L/day	Average daily consumption of drinking water by an adult
Exposure frequency	EF	365	days/year	Number of days an individual is exposed to contaminated water annually
Exposure duration	ED	70	years	Average lifetime exposure duration assumed for adults
Body weight	BW	70	kg	Average body weight of an adult
averaging time	AT	25,550	days	Lifetime exposure period calculated as ED × 365
Reference dose (Cu)	RfD	0.04	mg/kg/day	Oral reference dose for copper
Reference dose (Zn)	RfD	0.30	mg/kg/day	Oral reference dose for zinc
Reference dose (Pb)	RfD	0.0035	mg/kg/day	Oral reference dose for lead
Reference dose (Ni)	RfD	0.02	mg/kg/day	Oral reference dose for nickel
Reference Dose (CO)	RfD	0.0003	mg/kg/day	Oral reference dose for cobalt
Reference dose (Fe)	RfD	0.70	mg/kg/day	Oral reference dose for iron
Reference dose (Cd)	RfD	0.001	mg/kg/day	Oral reference dose for cadmium

The parameters were adapted to reflect Nigerian population exposure scenarios. The ingestion rate (2.0 L/day) represents average adult water consumption under tropical conditions. The body weight (65 kg) reflects regional anthropometric estimates for Nigerian adults. The exposure duration (54 years) was based on Nigeria's average life expectancy, ensuring a realistic lifetime risk assessment framework. For magnesium (Mg), reference dose values were derived from recommended daily intakes of 420 mg (men), 320 mg (women), and 80 mg for children (15 kg body weight), consistent with dietary exposure guidelines

2.10 Statistical analysis

Data obtained from physicochemical, bacteriological, heavy metal, and antibiotic susceptibility analyses were subjected to statistical analysis using IBM SPSS Statistics version 24. Descriptive statistics were expressed as mean ± standard deviation. One-way analysis of variance (ANOVA) followed by Duncan's multiple range post hoc test was used to determine significant differences among the sampling locations, with statistical significance set at $p < 0.05$. Pearson correlation analysis was performed to evaluate relationships among physicochemical and heavy metal parameters. Contamination Factor (CF), Heavy Metal Pollution Index (HPI), Health Risk Index (HRI), and Multiple Antibiotic Resistance (MAR) index were calculated using established equations to assess contamination levels, potential health risks, and antibiotic resistance patterns of the bacterial isolates.

3 Results and discussion

3.1 Results

3.1.1 The physicochemical properties of different sachet water samples sold (February, 2026)

The physicochemical characteristics of the water samples collected in February, 2026 showed moderate variation among the sampling locations (Table 2). The pH values ranged from 5.80 ± 0.10 in W1 (lowest) to 7.37 ± 0.15 in W2 (highest), indicating that most samples were slightly acidic to near neutral, with W2 and W5 (7.20 ± 0.10) falling within the recommended range. Electrical conductivity (EC) varied between 0.20 ± 0.00 mS/cm in W5 (lowest) and 1.20 ± 0.00 μS/cm in W3 (highest). Similarly, total dissolved solids (TDS) recorded the highest value in W3 (600.00 ± 0.00 ppm) and the lowest values

Table 2 Physicochemical characteristics of different sachet water samples sold (February, 2026)

Parameter	W1	W2	W3	W4	W5	W6	W7	W8	F-value	p-value	WHO Standard
Colour	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	–	–	Unobjectionable
Odour	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	–	–	Unobjectionable
Taste	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	–	–	Unobjectionable
pH	5.80 ± 0.10 ^d	7.37 ± 0.15 ^a	6.50 ± 0.10 ^b	6.37 ± 0.23 ^c	7.20 ± 0.10 ^a	6.67 ± 0.15 ^b	6.10 ± 0.10 ^c	6.47 ± 0.06 ^b	41.26	< 0.001	6.5–8.5
EC (mS/cm)	0.32 ± 0.03 ^d	0.70 ± 0.00 ^b	1.20 ± 0.00 ^a	0.60 ± 0.00 ^c	0.20 ± 0.00 ^c	0.70 ± 0.00 ^b	0.47 ± 0.12 ^{ad}	0.50 ± 0.00 ^c	186.44	< 0.001	1000
TDS (ppm)	100.00 ± 0.00 ^c	300.00 ± 0.00 ^b	600.00 ± 0.00 ^a	300.00 ± 0.00 ^b	100.00 ± 0.00 ^c	270.00 ± 0.06 ^b	100.00 ± 0.00 ^c	100.00 ± 0.00 ^a	212.38	< 0.001	500
Temperature (°C)	27.33 ± 0.58 ^b	28.00 ± 0.00 ^a	28.00 ± 0.00 ^a	28.00 ± 0.00 ^a	27.67 ± 0.58 ^b	28.00 ± 0.00 ^a	28.00 ± 0.00 ^a	28.00 ± 0.00 ^a	3.57	0.028	Ambient
Turbidity (NTU)	1.00 ± 0.09 ^c	1.40 ± 0.12 ^b	1.90 ± 0.15 ^a	1.20 ± 0.14 ^b	0.90 ± 0.08 ^c	1.30 ± 0.10 ^b	1.10 ± 0.12 ^b	1.70 ± 0.13 ^a	6.41	0.012	≤ 5
DO (mg/L)	6.10 ± 0.14 ^b	6.80 ± 0.20 ^a	6.20 ± 0.15 ^b	6.90 ± 0.21 ^a	7.20 ± 0.16 ^a	6.70 ± 0.18 ^a	6.50 ± 0.17 ^b	6.40 ± 0.18 ^b	5.32	0.019	≥ 5
BOD (mg/L)	1.50 ± 0.07 ^a	1.30 ± 0.05 ^b	1.80 ± 0.07 ^a	1.10 ± 0.05 ^b	0.90 ± 0.04 ^c	1.20 ± 0.06 ^b	1.30 ± 0.05 ^b	1.70 ± 0.06 ^a	7.84	0.010	≤ 3
COD (mg/L)	7.00 ± 0.33 ^a	6.20 ± 0.30 ^b	8.50 ± 0.40 ^a	5.80 ± 0.27 ^b	4.90 ± 0.25 ^c	6.00 ± 0.28 ^b	6.30 ± 0.29 ^b	8.00 ± 0.35 ^a	9.11	0.006	≤ 10
TOD (mg/L)	9.60 ± 0.42 ^a	8.90 ± 0.40 ^b	11.50 ± 0.50 ^a	8.20 ± 0.36 ^b	7.00 ± 0.30 ^c	8.50 ± 0.38 ^b	9.00 ± 0.39 ^b	10.90 ± 0.45 ^a	7.23	0.011	–
H ₂ S (mg/L)	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.01 ± 0.00 ^b	0.01 ± 0.00 ^b	0.01 ± 0.00 ^b	0.01 ± 0.00 ^b	0.02 ± 0.01 ^a	2.02	0.150	≤ 0.05

NB: Different superscript letters in the same column indicate statistically significant differences among sample means (one-way ANOVA followed by post-hoc test—Duncan's multiple range test at $p < 0.05$). Means with the same letter are not significantly different

pH Potential of Hydrogen (measure of acidity or alkalinity of water), EC Electrical Conductivity, $\mu\text{S}/\text{cm}$ Microsiemens per centimeter (unit of electrical conductivity), TDS Total Dissolved Solids, ppm Parts per million, Temp Temperature, °C Degree Celsius, DO Dissolved Oxygen, BOD Biochemical Oxygen Demand, COD Chemical Oxygen Demand, TOD Total Oxygen Demand, H₂S Hydrogen Sulfide, NTU Nephelometric Turbidity Units (unit of turbidity), Mean ± Standard Deviation, WHO World Health Organization

in W5, W1, and W4 (100.00 ± 0.00 ppm). Temperature remained fairly constant, with the highest value (28.00°C) observed in most samples and the lowest in W1 ($27.33 \pm 0.58^\circ\text{C}$). Turbidity was highest in W3 (1.90 NTU) and lowest in W5 (0.90 NTU). Dissolved oxygen (DO) peaked in W5 (7.20 mg/L) and was lowest in W1 (6.10 mg/L). BOD, COD, and TOD were highest in W3 (1.80 mg/L, 8.50 mg/L, and 11.50 mg/L respectively) and lowest in W5 (0.90 mg/L, 4.90 mg/L, and 7.00 mg/L). Hydrogen sulfide (H_2S) concentrations were minimal, ranging from 0.01 mg/L in W6, W4, W5, and W4 to 0.02 mg/L in W2, W3, W1, and W9, suggesting negligible sulphide contamination in the water samples.

3.1.2 Total bacterial count of different sachet water samples

The total bacterial counts of the sachet water samples showed considerable variation among the samples (Table 3). The total heterotrophic count (THC) ranged from 0.00 ± 0.00 CFU/ml in SOA-4 (lowest) to 110.00 ± 5.00 CFU/ml in SOA-5 (highest), exceeding the recommended limit of 100 CFU/ml. Moderate counts were observed in SOA-3 (17.00 ± 1.50 CFU/ml), while SOA-7 (2.00 ± 0.20 CFU/ml) and SOA-1 and SOA-8 (1.00 CFU/ml) recorded relatively low values. SOA-2 (0.22 ± 0.05 CFU/ml) and SOA-6 (0.50 ± 0.08 CFU/ml) also showed minimal bacterial presence. For coliform counts, the highest value occurred in SOA-3 (6.00 ± 0.50 CFU/ml), whereas SOA-2, SOA-4, SOA-5, and SOA-7 recorded zero coliforms.

3.1.3 Cultural morphology of the *Escherichia coli* isolate from sachet water samples

Sachet water isolates exhibited noticeable morphological diversity, though most characteristics were consistent with presumptive *Escherichia coli* colonial isolates (Table 4). Most isolates were milky in colour with pulvinate or raised elevation, predominantly circular or round with entire margins, indicating typical *Escherichia coli* morphology. Slight variations were observed in SOA-IS3 and SOA-IS8, which showed purple pigmentation, filamentous shape, curled margins, and brittle matte texture, suggesting atypical phenotypic expression or possible mixed culture contamination within sachet water samples.

3.1.4 Biochemical characterisation of *Escherichia coli* isolates from sachet water samples

The biochemical characterization of the nine isolates from sachet water samples revealed a consistent profile typical of *Escherichia coli* (Table 5). All isolates were Gram-negative

Table 3 Total bacterial count of different sachet water samples sold

Sample code	Total heterotrophic count (CFU/ml)	Coliform count (CFU/ml)
SOA-1	1.00 ± 0.10^c	1.00 ± 0.10^b
SOA-2	0.22 ± 0.05^d	ND
SOA-3	17.00 ± 1.50^b	6.00 ± 0.50^a
SOA-4	ND	ND
SOA-5	110.00 ± 5.00^a	ND
SOA-6	0.50 ± 0.08^d	0.40 ± 0.05^b
SOA-7	2.00 ± 0.20^c	ND
SOA-8	1.00 ± 0.12^c	0.10 ± 0.02^b
WHO Limits	100	0
F-value	84.63	39.42
p-value	< 0.001	< 0.001

Values are Mean $\pm$ Standard Deviation ($n=3$); different superscript letters within the same column indicate significant differences ($p < 0.05$)

PCA Plate Count Agar, EMB Eosin Methylene Blue Agar, WHO World Health Organisation, ND Not Detected

Table 4 Cultural morphology of the *Escherichia coli* isolate from sachet water samples

Isolate Code	Colour	Elevation	Texture	Shape	Margin
SOA-IS1	White	Pulvinate	Shiny	Circular	Entire
SOA-IS2	Milky	Raised	Dry, Mucoid	Round	Entire
SOA-IS3	Purple	Pulvinate	Matte, Brittle	Filamentous	Curled
SOA-IS4	Milky	Pulvinate	Translucent	Round	Entire
SOA-IS5	Milky	Raised	Dry, Mucoid	Round	Entire
SOA-IS6	Milky	Pulvinate	Dry, Mucoid	Round	Entire
SOA-IS7	Milky	Raised	Dry, Mucoid	Round	Entire
SOA-IS8	Purple	Pulvinate	Matte, Brittle	Filamentous	Curled
SOA-IS9	Milky	Pulvinate	Translucent	Round	Entire

rods and showed positive (+ve) reactions for indole, methyl red, catalase, and motility tests, while testing negative (–ve) for Voges–Proskauer, citrate utilization, urease, and oxidase. The Triple Sugar Iron (TSI) reaction showed acid/acid (A/A) with gas production but no hydrogen sulphide formation, which is a characteristic metabolic pattern of *Escherichia coli*.

3.1.5 Antibiotic susceptibility pattern of *Escherichia coli* isolates based on zone of inhibition in (mm)

The antibiotic susceptibility pattern of *Escherichia coli* isolates revealed marked variability in response to the tested antibiotics (Table 6). All isolates were highly susceptible to ofloxacin, with inhibition zones ranging from 26.00 to 34.00 mm (S), while cefoperazone also showed good activity in most isolates (30.00–36.00 mm), except SOA-IS3 and SOA-IS9 which recorded low zones of 10.00 mm and 12.00 mm (R). Nitrofurantoin demonstrated selective effectiveness, with high susceptibility in SOA-IS2, SOA-IS4, SOA-IS6, and SOA-IS8 (34.00 mm), but resistance in others (8.00–12.00 mm). Complete resistance (8.00 mm) was observed across all isolates for augmentin, ampicillin, ceftazidime, cefuroxime, and gentamicin, except slight inhibition in gentamicin (14.00 mm) for SOA-IS1, IS5, and IS7.

3.1.6 Multiple antibiotic resistance (MAR) index and percentage resistance of bacterial isolates

The multiple antibiotic resistance (MAR) index results indicate a high level of resistance among the bacterial isolates (Table 7). The MAR values ranged from 0.63 to 0.75, corresponding to resistance percentages of 62.5% to 75.0%. Isolates SOA-IS1, SOA-IS3, SOA-IS5, SOA-IS7, and SOA-IS9 exhibited the highest resistance, each resisting 6 out of 8 antibiotics (MAR = 0.75; 75.0%), including augmentin, nitrofurantoin, ampicillin, ceftazidime, cefuroxime, and gentamicin. In contrast, SOA-IS2, SOA-IS4, SOA-IS6, and SOA-IS8 showed slightly lower resistance, each resisting 5 antibiotics (MAR = 0.63; 62.5%), mainly excluding nitrofurantoin. All isolates had MAR values above 0.2.

3.1.7 Heavy metal analysis of different sachet water samples sold

The heavy metal concentrations in the water samples showed slight variations across the sampling points (Table 8). Copper (Cu) concentrations ranged from 0.010 ± 0.001 ppm in W2, W6, W4, W5, and the second W4 (lowest) to 0.020 ± 0.002 ppm in W3, W1, and W9 (highest). Zinc (Zn) recorded the highest value in W3 (0.050 ± 0.003 ppm) and the lowest in W5 (0.020 ± 0.002 ppm). Lead (Pb) concentrations were highest in W3, W4, W1, and W9 (0.006 ± 0.001 ppm), while the lowest value occurred in W5 (0.004 ± 0.001 ppm).

Table 5 Biochemical characterisation of *Escherichia coli* isolates from sachet water samples

S/n	Isolate Code	Gram Reaction	Indole	Methyl Red	Voges-Proskauer	Citrate Utilization	Urease	Oxidase	Catalase	TSI Reaction	Motility	Organism
1	SOA-IS1	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>
2	SOA-IS2	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>
3	SOA-IS3	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>
4	SOA-IS4	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>
5	SOA-IS5	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>
6	SOA-IS6	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>
7	SOA-IS7	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>
8	SOA-IS8	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>
9	SOA-IS9	GNR	+	+	-	-	-	-	+	A/A, Gas+, H ₂ S-	+	<i>Escherichia coli</i>

GNR Gram Negative Rod, - Negative, + Positive, TSI Acid/Acid with gas production and no hydrogen sulfide

Table 6 Antibiotic susceptibility pattern of *Escherichia coli* isolates based on zone of inhibition in (mm)

Isolate Code	Ofloxacin (OPL)	Augmentin (AUG)	Cefoperazone (CPZ)	Nitrofurantoin (NIT)	Ampicillin (AMP)	Ceftazidime (CAZ)	Cefuroxime (CRX)	Gentamicin (GEN)
SOA-IS1	26.00 (S)	8.00 (R)	30.00 (S)	12.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)	14.00 (R)
SOA-IS2	34.00 (S)	8.00 (R)	36.00 (S)	34.00 (S)	8.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)
SOA-IS3	28.00 (S)	8.00 (R)	10.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)
SOA-IS4	34.00 (S)	8.00 (R)	36.00 (S)	34.00 (S)	8.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)
SOA-IS5	26.00 (S)	8.00 (R)	30.00 (S)	12.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)	14.00 (R)
SOA-IS6	34.00 (S)	8.00 (R)	36.00 (S)	34.00 (S)	8.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)
SOA-IS7	26.00 (S)	8.00 (R)	30.00 (S)	12.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)	14.00 (R)
SOA-IS8	34.00 (S)	8.00 (R)	36.00 (S)	34.00 (S)	8.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)
SOA-IS9	28.00 (S)	8.00 (R)	12.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)	8.00 (R)

S Susceptible, R Resistant

Table 7 Multiple antibiotic resistance (MAR) index and percentage resistance of bacterial isolates

Isolate code	Antibiotics resistant to	No. of antibiotics resistant (a)	Total antibiotics tested (b)	MAR index (a/b)	MAR (%)
SOA-IS1	Augmentin, Nitrofurantoin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	6	8	0.75	75.0
SOA-IS2	Augmentin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	5	8	0.63	62.5
SOA-IS3	Augmentin, Nitrofurantoin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	6	8	0.75	75.0
SOA-IS4	Augmentin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	5	8	0.63	62.5
SOA-IS5	Augmentin, Nitrofurantoin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	6	8	0.75	75.0
SOA-IS6	Augmentin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	5	8	0.63	62.5
SOA-IS7	Augmentin, Nitrofurantoin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	6	8	0.75	75.0
SOA-IS8	Augmentin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	5	8	0.63	62.5
SOA-IS9	Augmentin, Nitrofurantoin, Ampicillin, Ceftazidime, Cefuroxime, Gentamicin	6	8	0.75	75.0

Nickel (Ni) ranged from 0.003 ± 0.001 ppm in W5 (lowest) to 0.005 ± 0.001 ppm in W3, W1, and W9 (highest). Cobalt (Co) varied between 0.001 ± 0.000 ppm in W5 (lowest) and 0.003 ± 0.001 ppm in W3 and W9 (highest). Iron (Fe) showed the highest concentration in W1 (0.160 ± 0.010 ppm) and the lowest in W5 (0.100 ± 0.010 ppm). Cadmium (Cd) remained constant across all samples at 0.001 ± 0.000 ppm.

3.1.8 Pearson correlation matrix of physicochemical parameters and heavy metals

The Pearson correlation matrix revealed significant relationships among the physicochemical parameters and heavy metals in the sachet water samples (Table 9). Strong positive correlations were observed among BOD, COD, and TOD ($r = 0.99$ – 1.00), indicating that these parameters are closely associated. Turbidity also showed strong positive relationships with EC and TDS ($r = 0.82$ – 0.95), suggesting that increased suspended and dissolved solids contribute to higher turbidity levels. Dissolved oxygen (DO) exhibited strong negative correlations with BOD, COD, Ni, and Fe ($r = -0.85$ to -0.93), implying

Table 8 Heavy metal concentrations of different sachet water samples sold (February, 2026)

Sample	Cu (ppm)	Zn (ppm)	Pb (ppm)	Ni (ppm)	Co (ppm)	Fe (ppm)	Cd (ppm)
W1	0.020 ± 0.002 ^a	0.030 ± 0.002 ^c	0.006 ± 0.001 ^a	0.005 ± 0.001 ^a	0.002 ± 0.000 ^b	0.160 ± 0.010 ^a	0.001 ± 0.000 ^a
W2	0.010 ± 0.001 ^b	0.030 ± 0.002 ^c	0.005 ± 0.001 ^b	0.004 ± 0.001 ^b	0.002 ± 0.000 ^b	0.120 ± 0.010 ^c	0.001 ± 0.000 ^a
W3	0.020 ± 0.002 ^a	0.050 ± 0.003 ^a	0.006 ± 0.001 ^a	0.005 ± 0.001 ^a	0.003 ± 0.001 ^a	0.150 ± 0.010 ^a	0.001 ± 0.000 ^a
W4	0.010 ± 0.001 ^b	0.040 ± 0.002 ^b	0.006 ± 0.001 ^a	0.004 ± 0.001 ^b	0.002 ± 0.000 ^b	0.140 ± 0.010 ^b	0.001 ± 0.000 ^a
W5	0.010 ± 0.001 ^b	0.020 ± 0.002 ^d	0.004 ± 0.001 ^c	0.003 ± 0.001 ^c	0.001 ± 0.000 ^c	0.100 ± 0.010 ^d	0.001 ± 0.000 ^a
W6	0.010 ± 0.001 ^b	0.030 ± 0.002 ^c	0.005 ± 0.001 ^b	0.004 ± 0.001 ^b	0.002 ± 0.000 ^b	0.130 ± 0.010 ^b	0.001 ± 0.000 ^a
W7	0.010 ± 0.001 ^b	0.030 ± 0.002 ^c	0.005 ± 0.001 ^b	0.004 ± 0.001 ^b	0.002 ± 0.000 ^b	0.140 ± 0.010 ^b	0.001 ± 0.000 ^a
W8	0.020 ± 0.002 ^a	0.040 ± 0.002 ^b	0.006 ± 0.001 ^a	0.005 ± 0.001 ^a	0.003 ± 0.001 ^a	0.150 ± 0.010 ^a	0.001 ± 0.000 ^a
WHO Standard	2.00	3.00	0.01	0.07	–	0.300	0.50
F-value	18.42	46.37	12.54	15.61	9.74	22.88	0.00
p-value	< 0.001	< 0.001	0.002	< 0.001	0.004	< 0.001	1.000

NB: Values are Mean ± Standard Deviation (n = 3); Different superscript letters within a column indicate significant difference (*p* < 0.05)
± Standard deviation, Cu Copper, Zn Zinc, Pb Lead, Ni Nickel, Co Cobalt, Fe Iron, Cd Cadmium, ppm part per million

Table 9 Pearson correlation matrix of physicochemical parameters and heavy metals

Parameter	pH	EC	TDS	Temperature	Turbidity	DO	BOD	COD	TOD	H ₂ S	Cu	Zn	Pb	Ni	Co	Fe
pH	1.000	0.073	0.167	0.327	0.073	0.714	-0.404	-0.398	-0.383	-0.051	-0.480	-0.310	-0.639	-0.599	-0.337	-0.847
EC	0.073	1.000	0.831	0.579	0.823	-0.352	0.560	0.597	0.617	0.330	0.238	0.792	0.419	0.431	0.672	0.294
TDS	0.167	0.831	1.000	0.608	0.953	-0.265	0.629	0.685	0.689	0.472	0.412	0.857	0.543	0.516	0.805	0.301
Temperature	0.327	0.579	0.608	1.000	0.577	0.213	0.091	0.136	0.181	-0.183	-0.327	0.442	0.027	-0.073	0.408	-0.118
Turbidity	0.073	0.823	0.953	0.577	1.000	-0.448	0.790	0.826	0.837	0.583	0.531	0.844	0.537	0.631	0.899	0.397
DO	0.714	-0.352	-0.265	0.213	-0.448	1.000	-0.880	-0.848	-0.846	-0.650	-0.820	-0.547	-0.726	-0.927	-0.722	-0.903
BOD	-0.404	0.560	0.629	0.091	0.790	-0.880	1.000	0.993	0.994	0.795	0.867	0.748	0.730	0.935	0.922	0.774
COD	-0.398	0.597	0.685	0.136	0.826	-0.848	0.993	1.000	0.997	0.754	0.869	0.806	0.750	0.923	0.941	0.767
TOD	-0.383	0.617	0.689	0.181	0.837	-0.846	0.994	0.997	1.000	0.754	0.835	0.804	0.739	0.917	0.950	0.761
H ₂ S	-0.051	0.330	0.472	-0.183	0.583	-0.650	0.795	0.754	0.754	1.000	0.775	0.438	0.539	0.756	0.626	0.487
Cu	-0.480	0.238	0.412	-0.327	0.531	-0.820	0.867	0.869	0.835	0.775	1.000	0.565	0.696	0.878	0.700	0.736
Zn	-0.310	0.792	0.857	0.442	0.844	-0.547	0.748	0.806	0.804	0.438	0.565	1.000	0.812	0.717	0.882	0.659
Pb	-0.639	0.419	0.543	0.027	0.537	-0.726	0.730	0.750	0.739	0.539	0.696	0.812	1.000	0.882	0.786	0.911
Ni	-0.599	0.431	0.516	-0.073	0.631	-0.927	0.935	0.923	0.917	0.756	0.878	0.717	0.882	1.000	0.867	0.919
Co	-0.337	0.672	0.805	0.408	0.899	-0.722	0.922	0.941	0.950	0.626	0.700	0.882	0.786	0.867	1.000	0.739
Fe	-0.847	0.294	0.301	-0.118	0.397	-0.903	0.774	0.767	0.761	0.487	0.736	0.659	0.911	0.919	0.739	1.000

Cu Copper, Zn Zinc, Pb Lead, Ni Nickel, Co Cobalt, Fe Iron, pH Potential of Hydrogen (measure of acidity or alkalinity of water), EC Electrical Conductivity, TDS Total Dissolved Solids, DO Dissolved Oxygen, BOD Biochemical Oxygen Demand, COD Chemical Oxygen Demand, TOD Total Oxygen Demand, H₂S Hydrogen Sulfide

Table 10 Contamination factor ($CF = C_m/C_s$) per sampling site per metal

Sample code	CF(Cu)	CF(Zn)	CF(Pb)	CF(Ni)	CF(Fe)	CF(Cd)	Category
W1	0.010	0.010	0.600	0.071	0.533	0.002	Low
W2	0.005	0.010	0.500	0.057	0.400	0.002	Low
W3	0.010	0.016	0.600	0.071	0.500	0.002	Low
W4	0.005	0.013	0.600	0.057	0.466	0.002	Low
W5	0.005	0.006	0.400	0.042	0.333	0.002	Low
W6	0.005	0.010	0.500	0.057	0.433	0.002	Low
W7	0.005	0.010	0.500	0.057	0.466	0.002	Low
W8	0.010	0.013	0.600	0.071	0.500	0.002	Low

Co excluded; (no WHO standard). $CF < 1$ = low; 1–3 = moderate; 3–6 = considerable; ≥ 6 = very high

Table 11 Heavy metal pollution index (HPI) per sampling site

Sample code	HPI value	Classification
W1	52.145	Low pollution
W2	43.303	Low pollution
W3	52.055	Low pollution
W4	51.790	Low pollution
W5	34.646	Low pollution
W6	43.395	Low pollution
W7	43.487	Low pollution
W8	52.054	Low pollution

$HPI = \sum (W_i \times Q_i) \times 100$, where $Q_i = C_i/S_i$ and $W_i = (1/S_i)/\sum(1/S_i)$. Threshold: $HPI < 100$ = low pollution

that higher organic load and metal concentrations may reduce oxygen availability due to microbial oxidation processes. Additionally, heavy metals such as Pb, Ni, Co, and Fe demonstrated strong positive correlations ($r = 0.74$ – 0.92). The negative correlation between pH and Fe ($r = -0.85$).

3.1.9 Potential risk assessment analysis

3.1.9.1 Contamination factor (CF) All metals at every sampling site returned CF values strictly less than 1, placing every site-metal combination in the Low contamination category (Table 10). The highest individual CF values were recorded for Lead at W1, W3, W4, W7 and W8 ($CF = 0.600$) and Iron at W1 ($CF = 0.533$), confirming those two metals as the closest to their permissible thresholds. Cadmium registered the lowest CF across all sites ($CF = 0.002$), reflecting its extreme distance from the WHO limit of 0.5 ppm.

3.1.9.2 Heavy metal pollution index (HPI) The HPI for all eight location ranged from approximately 1.72 (W5) to 2.84 (W1), all dramatically below the critical threshold of 100 (Table 11). This unambiguously classifies every sampling point as exhibiting low overall heavy metal pollution. The HPI is dominated by the metals with the highest unit weight-ages ($W_i = 1/S_i$), meaning Lead with its very low WHO standard of 0.01 ppm carries the highest weight ($W_i \approx 0.901$ of the normalised total) and drives most of the inter-site HPI variation. W1 ranks highest due to its elevated Fe and Pb concentrations co-occurring.

3.1.9.3 Possible health risk index (HRI) The Health Risk Index (HRI) assessment indicated that all water samples posed no significant non-carcinogenic health risk to consumers, as the total HRI values ranged from 0.159 (W5) to 0.220 (W3), all of which were below the critical threshold of 1.0 (Table 12). Among the analyzed metals, lead (Pb) contributed the highest HRI values (0.081–0.122), making it the major contributor to the

Table 12 Possible Health Risk Index (HRI) per sampling site per metal

Sample code	HRI(Cu)	HRI(Zn)	HRI(Pb)	HRI(Ni)	HRI(Co)	HRI(Fe)	HRI(Cd)	Total HRI	Status
W1	0.014	0.002	0.122	0.007	0.005	0.006	0.057	0.216	Acceptable
W2	0.007	0.002	0.102	0.005	0.005	0.004	0.057	0.185	Acceptable
W3	0.014	0.004	0.122	0.007	0.008	0.006	0.057	0.220	Acceptable
W4	0.007	0.003	0.122	0.005	0.005	0.005	0.057	0.207	Acceptable
W5	0.007	0.001	0.081	0.004	0.002	0.004	0.057	0.159	Acceptable
W6	0.007	0.002	0.102	0.005	0.005	0.005	0.057	0.185	Acceptable
W7	0.007	0.002	0.102	0.005	0.005	0.005	0.057	0.186	Acceptable
W8	0.014	0.003	0.122	0.007	0.008	0.006	0.057	0.219	Acceptable

HRI = $(C \times IR \times EF \times ED)/(BW \times AT \times RfD)$. IR = 2 L/d, EF = 365d/yr, ED = 70 year, BW = 70 kg, AT = 25550d. HRI ≥ 1 = potential risk

overall health risk, followed by cadmium (Cd) with a constant HRI of 0.057 across all sampling sites. Conversely, zinc (Zn), cobalt (Co), nickel (Ni), copper (Cu), and iron (Fe) contributed relatively low HRI values.

3.2 Discussion

The present study provides a comprehensive evaluation of the physicochemical quality, bacteriological status, antibiotic resistance profile, and heavy metal risk of sachet water, revealing important insights into its safety for human consumption within the communities concerned. Sachet water remains one of the most widely consumed sources of drinking water in many developing countries because of its affordability and accessibility, especially in urban and peri-urban communities [8]. However, concerns regarding production standards, storage conditions, bacterial contamination, and environmental pollutants have continued to raise questions about its public health safety [51]. The findings of this study therefore contribute valuable evidence on the bacteriological and chemical integrity of sachet water sold within the study area, while also highlighting emerging antibiotic resistance concerns associated with contaminated drinking water sources. Similar broad concerns on packaged drinking water quality have been discussed by [29].

In this current study, the physicochemical characteristics of the sachet water samples indicate generally acceptable water quality with slight spatial variability. The pH range of 5.80–7.37 suggests mildly acidic to neutral conditions and falls close to the acceptable guideline range for potable water. Samples W2 and W5 were near neutrality, indicating better buffering capacity and lower acidity than W1. These findings are comparable to those reported by [29], who observed similar pH fluctuations in packaged water brands sold within the Samaru community, Northwest Nigeria. The relatively low electrical conductivity (0.20–1.20 $\mu\text{S}/\text{cm}$) and total dissolved solids (0.10–0.50 ppm) values reflect minimal ionic and mineral content, which is characteristic of treated or filtered drinking water. Such low dissolved solids may indicate effective purification processes, including reverse osmosis or demineralization steps often used in sachet water production [52]. Comparable observations were documented by [53] in treated drinking water systems.

Turbidity values were generally low, although W3 recorded the highest value of 1.90 NTU. While still within acceptable limits, the elevated turbidity in W3 compared with other samples may indicate suspended particles introduced during packaging, transport, or storage. Organic load indicators such as biochemical oxygen demand (1.80 mg/L), chemical oxygen demand (8.50 mg/L), and total oxygen demand (11.50 mg/L) were also highest in W3, suggesting localized contamination or higher biodegradable matter in that sample. Dissolved oxygen ranged from 6.10 to 7.20 mg/L, with the lowest value

observed in W1 and the highest in W5. The inverse relationship between dissolved oxygen and organic parameters supports microbial oxygen utilization and decomposition processes, where higher organic matter stimulates microbial respiration and reduces oxygen levels [54] and this trend agrees with the water quality dynamics described by [55]. The physicochemical data indicate that most samples were chemically safe, though some variation suggests the need for tighter quality consistency across brands.

Bacteriological analysis revealed variable contamination levels among the sachet water samples. Total heterotrophic bacterial counts ranged from 0.00 to 110.00 CFU/ml, with SOA-5 exceeding the recommended permissible limit of 100 CFU/mL. Elevated heterotrophic counts in packaged water often indicate inadequate treatment, poor sanitation during processing, contaminated filling machines, or unhygienic storage environments. Similar trends have been reported in sachet water studies across Nigeria, supporting the findings of [9]. The presence of coliform bacteria, with counts reaching 6.00 CFU/mL, further suggests possible faecal contamination or post-treatment contamination. Coliform detection in drinking water is a key sanitary indicator because these organisms are associated with contamination pathways that may also introduce enteric pathogens [56]. This aligns with warnings by the World Health Organization on the risks associated with inadequately treated drinking water [57].

The predominance of *Escherichia coli* based on colonial morphology and biochemical characterization is particularly significant because *Escherichia coli* is a recognized indicator of faecal pollution. All isolates were Gram-negative rods, positive for indole, methyl red, catalase, and motility, while negative for citrate, urease, oxidase, and Voges–Proskauer reactions, consistent with standard identification features of *Escherichia coli*. Similar findings have been documented by [58, 59]. The detection of *Escherichia coli* in sachet water indicates serious sanitary lapses, as potable water should ideally be free of faecal coliforms. Even low-level contamination can create a pathway for gastrointestinal infections, especially among immunocompromised individuals, children, and the elderly [17].

The antibiotic susceptibility profile highlights a concerning resistance pattern among the recovered *Escherichia coli* isolates. High susceptibility to ofloxacin (26.00–34.00 mm) and cefoperazone (30.00–36.00 mm) indicates that these antibiotics remain effective treatment options against the tested isolates. Nitrofurantoin also showed selective activity in some isolates, particularly those with inhibition zones of 34.00 mm. These findings are consistent with studies by [60, 61]. However, widespread resistance was observed against augmentin, ampicillin, ceftazidime, cefuroxime, and gentamicin. This pattern reflects growing antimicrobial pressure in the environment, where misuse or overuse of antibiotics in human medicine, livestock production, and improper waste disposal may select for resistant strains. The observations also align with reports by [62] who linked cephalosporin use with increasing extended-spectrum beta-lactamase prevalence.

Comparable multidrug resistance trends have been reported in environmental *Escherichia coli* isolates, suggesting persistent selective pressure from antibiotic misuse. The Multiple Antibiotic Resistance (MAR) index values of 0.63–0.75 far exceeded the threshold of 0.2, indicating that these isolates originated from high-risk environments where antibiotics are frequently encountered. Such environments may include sewage systems, hospital effluents, animal waste runoff, and polluted surface waters. Similar conclusions

were drawn by the [63], as well as recent studies by [64, 65]. The presence of multidrug-resistant *Escherichia coli* in consumable water is a major public health concern because it limits treatment options if infections occur and may facilitate horizontal transfer of resistance genes to other pathogens [66].

This study also reported heavy metal analysis that revealed low concentrations of all tested metals, including copper, zinc, lead, nickel, cobalt, iron, and cadmium, with values below recommended safety limits. This contrasts with the findings of [67] who reported excessive heavy metal contamination in groundwater. The strong positive correlations among BOD, COD, and TOD ($r = 0.99\text{--}1.00$) indicate common organic pollution sources, while negative correlations between dissolved oxygen and some metals suggest oxidative or redox processes influencing water chemistry. Contamination factor values below 1, heavy metal pollution index values of 34.65–52.15, and health risk index values of 0.159–0.220 confirm minimal ecological and human health risks from metal exposure through sachet water consumption.

Nevertheless, lead contributed the greatest share of the risk burden despite its low concentration because of its low reference dose and known toxicity, as also emphasized by [68, 69]. The findings suggest that while the chemical quality of the sachet water was largely acceptable, microbiological contamination and antibiotic resistance remain significant public health concerns. Regulatory agencies and producers should therefore strengthen hygienic production practices, routine microbiological screening, proper storage systems, and antibacterial resistance surveillance to ensure safer sachet water for consumers.

4 Conclusion

This study demonstrated that the examined sachet water samples were generally within acceptable physicochemical and heavy metal quality limits, although slight variations existed among sampling locations. Parameters such as pH, turbidity, dissolved oxygen, conductivity, and total dissolved solids indicated relatively safe water quality, while copper, zinc, lead, nickel, cobalt, iron, and cadmium concentrations remained below recommended guideline values. The contamination factor (CF), heavy metal pollution index (HPI), and health risk index (HRI) further confirmed low contamination status and negligible non-carcinogenic risk to adult consumers. However, bacteriological assessment revealed notable concerns. Some samples exceeded recommended heterotrophic bacterial counts, and coliform contamination was detected in selected sachet water brands, indicating lapses in hygiene, treatment efficiency, or post-production handling. Morphological and biochemical characterisation identified the isolates as *Escherichia coli*, confirming faecal contamination and potential public health implications. Antibiotic susceptibility testing showed high sensitivity to ofloxacin and moderate activity of cefoperazone and nitrofurantoin, whereas widespread resistance was observed against augmentin, ampicillin, ceftazidime, cefuroxime, and gentamicin. Elevated MAR indices (0.63–0.75) indicated exposure of the isolates to high-risk environments with frequent antibiotic pressure. Although chemical safety was satisfactory, microbiological contamination and multidrug-resistant *Escherichia coli* pose significant health risks. Continuous regulatory monitoring, improved production hygiene, and antimicrobial resistance surveillance are therefore essential to safeguard public health. Future studies will employ membrane filtration, API 20E and molecular identification techniques. It

should incorporate a larger sample size, repeated seasonal sampling, and a broader range of brands to improve representativeness and strengthen risk assessment.

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Author contributions

OJS—Conceptualization, writing-original draft, data curation, visualization; JOA—conceptualization, validation, writing—review and editing; ODO—validation, writing- review and editing; AOO—validation, writing—review and editing; OAO—validation, writing—review and editing; OVO—data curation, writing- review and editing; AAO—data curation, writing- review and editing; OAA—validation, writing- review and editing; AOO—validation, writing—review and editing; IBO—conceptualization, writing-original draft, data curation, project administration, supervision.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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